Local Preference Policies in Public Procurement: Efficiency, Distributional Effects, and Spillovers

Evidence from Chile's Compra Ágil Reform

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Introduction

Motivation

[PLACEHOLDER — fill with motivation slide]

Suggested content:

- Public procurement is large: OECD governments spend 12–15% of GDP on procurement. Developing countries often more.
- Local preference policies are widespread: Buy American, EU local content rules, developing-country SME preferences.
- Tension: local preferences may protect domestic firms and create local jobs, but can reduce competition and raise costs for the government.
- Little causal evidence on the competition and welfare effects of these policies, especially in developing countries.
- Chilean Compra Ágil reform (Dec 2024) provides a sharp, quasi-random change in the scope of local preference.

Research Questions

Q1 Efficiency and distributional effects: What are the short-run effects of local preference policies on competition in public procurement?

- **Efficiency:** Number and composition of bidders; Government procurement costs (winning bid prices) $\rightarrow$ Do local entry gains offset non-local exclusion?
- **Distributional effects:** Local vs. non-local firms; SME vs. large firms

Q2 Heterogeneity: How do the effects of the policy vary by region, based on pre-existing market structure and exogenous characteristics?

- Heterogeneous by government demand, supply-side market thickness, distance from Santiago, and other characteristics
- Heterogeneity in efficiency and distributional implications of the policy

Q3 Cross-market spillovers: Do local preference policies for 30–100 UTM tenders cause them to redirect bids to untreated bands (100–500+ UTM), in their home regions or outside?

- **Mechanism:** firm-level capacity constraints linking treated and untreated markets across regions.
- **Outcomes in untreated bands:** entry, bidder composition, and winning prices.
- **Welfare:** do cross-market effects amplify or offset the direct efficiency and distributional effects in Q1?

This Project

Part I: Empirics

- Descriptive facts about local and non-local firm participation in Chilean public procurement
- DiD estimates of the short-run effects of the Compra Ágil reform on bidder entry, bidder composition, and winning prices
- Regional heterogeneity: how effects vary with pre-reform market structure and geography
- Cross-market spillovers: competition effects in *above*-threshold markets driven by firm capacity reallocation

Part II: Structural Model

- Spatial model of firm entry and bidding for public procurement projects
- Firms have home regions; face distance-based cost and entry cost disadvantages; are subject to capacity constraints linking markets
- Estimation using pre- and post-reform variation
- **Counterfactuals:**
  - Optimal local preference threshold by region
  - Welfare decomposition (cost vs. local benefit)
  - Alternative policy designs

Related Literature

[PLACEHOLDER — fill with related literature]

Key strands:

- **Public procurement and competition:** Athey, Levin & Seira (2011); Krasnokutskaya & Seim (2011); Decarolis (2014); Coviello & Mariniello (2014)
- **Local preference / Buy-national policies:** Marion (2007); Kang & Miller (2022); Shingal (2020); Buy American paper;
- **SME preferences in procurement:** Carril, Gonzalez-Lira & Walker (2022); Sorensen (2022)
- **Procurement reform in developing countries:** Bandiera et al. (2009); Best et al. (2017); Bosio et al. (2022)
- **Cross-market spillovers and capacity constraints:** Jeziorski & Krasnokutskaya (2016); Hoekmann et al., (2025)

Setting

Compra Ágil Reform (Ley 21.634)

Before (until Dec 11, 2024)

- Compra Ágil limit: **30 UTM** ($\approx$ CLP 2M, USD 2k)
- Projects 30–100 UTM went through competitive licitaciones
- No explicit SME/local priority at first stage
- Non-local firms competed freely in all markets

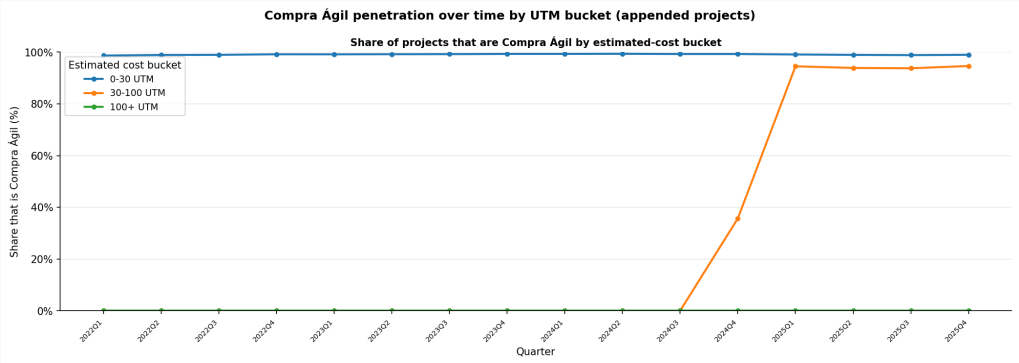
After (from Dec 12, 2024)

- Compra Ágil limit: **100 UTM** ($\approx$ CLP 7M, USD 7k)
- Projects 30–100 UTM now use simplified process
- **SME and local firm priority** in first instance
- Non-local firms effectively excluded from this band

Timeline: Law 21.634 published November 2024; effective December 12, 2024. First post-reform month with full exposure: January 2025.

Scale: The 30–100 UTM band accounts for $\approx$25–35% of all municipal procurement tenders by volume. The reform substantially narrows the set of markets where non-local firms can participate.

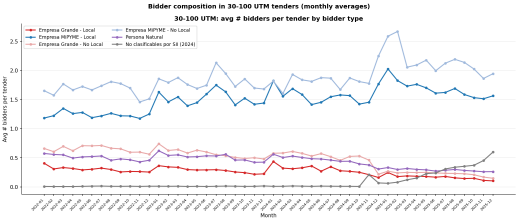
Projects Shifted Almost Entirely to Compra Ágil



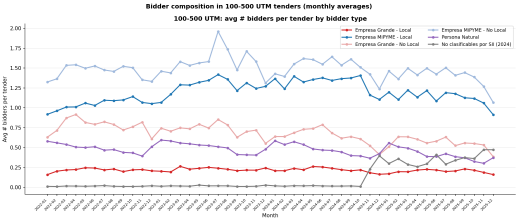
Takeaway: Nearly all 30–100 UTM projects switched from licitaciones to Compra Ágil after the reform — compliance is near-complete.

Bidder Composition by Band

30–100 UTM



100–500 UTM



Data

ChileCompra (Mercado Público)

- Universe of public procurement contracts in Chile
- Tender-level: estimated value (UTM), procurement type, region, sector, date
- Bid-level: firm RUT, bid amount, winner indicator
- **Coverage:** Jan 2022 – Mar 2025
- **Sample:** Municipalidades sector; treated band (30–100 UTM); control bands (0–30, 100–200 UTM)

Servicio de Impuestos Internos (SII)

- Chilean Internal Revenue Service administrative data
- Firm-level: size classification (micro, small, medium, large), sector, location, annual sales
- **Use:** Classify bidders as SME (MiPYME) vs. large; local vs. non-local (matched to buyer region)
- **Coverage:** Annual data, matched to ChileCompra via RUT

Final sample (Municipalidades): $\approx$2,500 procuring entities; $\approx$180,000 tenders; $\approx$650,000 bids over the study period.

Sample Overview: Pre-Reform Descriptives

[PLACEHOLDER — fill with summary statistics table]

Suggested content:

- Table: N tenders, N bidders (mean), SME win share, log price ratio — split by sector (Municipalidades vs. Obras Públicas) and UTM band (0–30, 30–100, 100–200 UTM), pre-reform only.
- Highlight that the treated band (30–100 UTM) had systematically lower bidder counts and higher non-local shares than control bands.
- Source: `did_tender_sample.parquet` + `did_bid_sample.parquet`.

Sample Overview: Bidder Characteristics (SII Data)

[PLACEHOLDER — fill with SII summary statistics]

Suggested content:

- Share of bids by SME vs. large firm, by UTM band and sector
- Share of bids by local vs. non-local firm (home region = buyer region)
- Average firm size (sales, employees) of winning vs. losing bidders
- Geographic concentration: fraction of bidders from RM bidding in other regions ("export share")

Empirical Strategy & Results

Descriptive Facts: Local vs. Non-Local Bidders

[PLACEHOLDER — fill with descriptive facts about local and non-local firm bids]

Suggested content:

- Bar/line chart: share of bids by local vs. non-local firms by UTM band and region, pre-reform
- Map or bar chart: which regions send firms to other regions most? (foreshadows the export share analysis)
- Heterogeneity: non-local share is higher in remote regions, lower in RM (the inverse-U pattern from the binscatter)

Difference-in-Differences: Specification

Estimating equation (TWFE):

$$y_{it} = \alpha_i + \gamma_t + \beta\, \mathbf{1}[\text{treated}_i \times \text{post}_t] + \varepsilon_{it}$$

Treatment and control groups (defined by estimated contract value in UTM):

- **Treated:** 30–100 UTM
  Tenders that switch from competitive bidding (licitación) to Compra Ágil after Dec 12, 2024.
- **Control 1** (low): 1–30 UTM
  Already under Compra Ágil before the reform.
- **Control 2** (high): 100–200 UTM
  Above the new threshold; procurement process unchanged.

Fixed effects: Procuring entity α_i + year-month γ_t.
Standard errors: Clustered by procuring entity.
Sample: Jan 2023 – Mar 2025; Municipalidades sector.

Results show control group: *Control 2 (100–200 UTM)*.

IV-DiD: Specification

Motivation: Intent-to-treat (DiD) estimates may under-state effects if compliance is imperfect. We instrument actual Compra Ágil use with reform eligibility.

First stage (within-entity, within-month):

$$\widetilde{CA}_{it} = \pi\, \widetilde{Z}_{it} + \nu_{it}, \qquad Z_{it} = \mathbf{1}[\text{treated}_i \times \text{post}_t]$$

Second stage:

$$\tilde{y}_{it} = \beta_{\text{IV}}\, \widehat{\widetilde{CA}}_{it} + \varepsilon_{it}$$

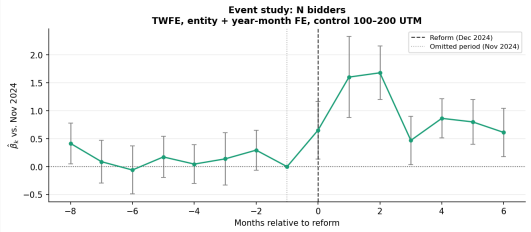
- **Endogenous:** CA_{it} — indicator, tender processed via Compra Ágil
- **Instrument:** $Z_{it} = \text{Treated}_i \times \text{Post}_t$ — reform eligibility
- Same FE and clustering as OLS DiD; Control 2 only (100–200 UTM)
- β_{IV} is a LATE: effect of Compra Ágil *use*

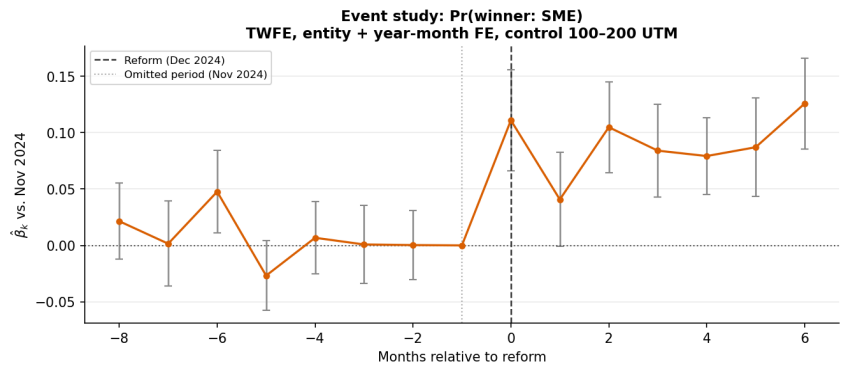
IV-DiD: First Stage (Municipalidades)

Sample	$\hat{\pi}$	F	N
Full tender sample	0.8854^{***}	9,710	234,893
	(0.0090)		
Winner sub-sample	0.8607^{***}	5,162	145,445
	(0.0120)		
Full tender (min bid)	0.8930^{***}	10,538	232,667
	(0.0087)		
Bid-level sample	0.8944^{***}	8,138	1,168,113
	(0.0099)		

$\hat{\pi}$ = OLS coef. of CA_{it} on Treated$_i$ $\times$ Post$_t$ (entity + year-month FE).
SE in parentheses, clustered by procuring entity. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.10$.
F = cluster-robust first-stage F-stat. Control: 100–200 UTM.

Event Studies: N Bidders and SME Share (Municipalidades)




Event study: Pr(winner: SME)
TWFE, entity + year-month FE, control 100–200 UTM
Reform (Dec 2024)
Omitted period (Nov 2024)
$\hat{\beta}_k$ vs. Nov 2024
0.15
0.10
0.05
0.00
−0.05
−8
−6
−4
−2
0
2
4
6
Months relative to reform

IV-DiD: Entry at Bidding Stage (Panel A, Municipalidades)

Outcome	OLS DiD: Control 100–200 UTM $\hat{\beta}_{\text{DiD}}$	N	IV-DiD: Control 100–200 UTM $\hat{\beta}_{\text{IV-DiD}}$	N
Panel A: Entry at Bidding Stage				
N bidders	0.8708^{***}	234,893	0.9834^{***}	234,893
	(0.0973)		(0.1096)	
N local bidders	0.3962^{***}	234,893	0.4474^{***}	234,893
	(0.0378)		(0.0421)	
N non-local bidders	0.5666^{***}	234,893	0.6399^{***}	234,893
	(0.0778)		(0.0880)	
N SME bidders	1.4797^{***}	234,893	1.6712^{***}	234,893
	(0.0712)		(0.0791)	
N large firm bidders	-0.5169^{***}	234,893	-0.5838^{***}	234,893
	(0.0396)		(0.0444)	
N non-SII bidders	-0.0920^{**}	234,893	-0.1039^{**}	234,893
	(0.0428)		(0.0484)	
Pr(single-bidder tender)	-0.0868^{***}	234,893	-0.0980^{***}	234,893
	(0.0076)		(0.0085)	

SE clustered by procuring entity. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.10$.

IV-DiD: Bidder Composition (Panel B, Municipalidades)

Outcome	OLS DiD: Control 100–200 UTM $\hat{\beta}_{\text{DiD}}$	N	IV-DiD: Control 100–200 UTM $\hat{\beta}_{\text{IV-DiD}}$	N
Panel B: Bidder Composition				
Any SME bidder	0.0265^{***}	217,461	0.0296^{***}	217,461
	(0.0031)		(0.0034)	
% bidders: SME	0.1024^{***}	217,461	0.1142^{***}	217,461
	(0.0048)		(0.0052)	
% bidders: large firm	-0.0958^{***}	217,461	-0.1069^{***}	217,461
	(0.0047)		(0.0051)	
% bidders: not in SII	-0.0966^{***}	234,893	-0.1091^{***}	234,893
	(0.0059)		(0.0065)	
% bidders: SME $\times$ local	-0.0182^{**}	217,461	-0.0203^{**}	217,461
	(0.0072)		(0.0081)	
% bidders: large $\times$ local	-0.0277^{***}	217,461	-0.0310^{***}	217,461
	(0.0031)		(0.0035)	
% bidders: non-SII $\times$ local	— — —	—	0.0000	217,461
	— — —		(0.0000)	

SE clustered by procuring entity. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.10$.

IV-DiD: Winner Characteristics (Panel C, Municipalidades)

Outcome	OLS DiD: Control 100–200 UTM $\hat{\beta}_{\text{DiD}}$	N	IV-DiD: Control 100–200 UTM $\hat{\beta}_{\text{IV-DiD}}$	N
Panel C: Winner Characteristics				
Pr(winner: SME)	0.0912***	145,445	0.1059***	145,445
	(0.0063)		(0.0070)	
Pr(winner: large firm)	−0.1353***	145,445	−0.1572***	145,445
	(0.0082)		(0.0093)	
Pr(winner: non-SII)	0.0008	177,903	0.0009	177,903
	(0.0068)		(0.0079)	
Pr(winner: SME & local)	0.0350***	145,444	0.0407***	145,444
	(0.0090)		(0.0104)	
Pr(winner: large & local)	−0.0368***	145,444	−0.0428***	145,444
	(0.0046)		(0.0053)	
Pr(winner: non-SII & local)	— — —	—	0.0000	145,444
	— — —		(0.0000)	

SE clustered by procuring entity. *** $p<0.01$, ** $p<0.05$, * $p<0.10$.

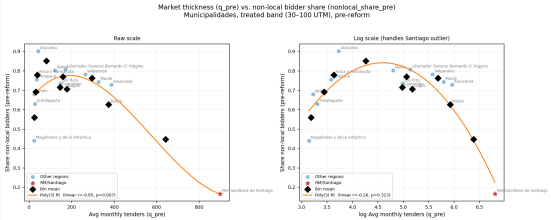
IV-DiD: Bid Outcomes (Panel D, Municipalidades)

Outcome	OLS DiD: Control 100–200 UTM $\hat{\beta}_{\text{DiD}}$	N	IV-DiD: Control 100–200 UTM $\hat{\beta}_{\text{IV-DiD}}$	N
Panel D: Bid Outcomes				
log(win bid/ref)	0.6748^{***}	179,165	0.7794^{***}	179,165
	(0.0444)		(0.0506)	
log(min bid/ref)	0.9736^{***}	232,667	1.0903^{***}	232,667
	(0.0683)		(0.0774)	
log(bid/ref)	0.6599^{***}	1,168,113	0.7378^{***}	1,168,113
	(0.0425)		(0.0477)	
Bid CV	-0.1132^{***}	183,756	-0.1261^{***}	183,756
	(0.0074)		(0.0084)	

SE clustered by procuring entity. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.10$.

Regional Heterogeneity

Regional Heterogeneity: Market Structure Varies by Region



Four pre-reform moderators (Municipalidades, 30–100 UTM):

- **Nonlocal share:** fraction of bids by out-of-region firms
- **q_pre:** avg. monthly tender count
- **n_pot_local:** unique local suppliers
- **Distance from Santiago:** geographic isolation (km)

Thick markets (high q_pre) have lower non-local penetration — market size proxies for local competitive capacity.

Regional Heterogeneity: Specification

Question: Does the effect of Compra Ágil vary with pre-reform regional characteristics?

Estimating equation (OLS DiD):

$$y_{irt} = \alpha_i + \gamma_t + \beta_1 \left(T_i \times \text{Post}_t \right) + \beta_2 \left(T_i \times \text{Post}_t \times Z_r \right) + \varepsilon_{irt}$$

- Z_r = standardised pre-reform regional moderator (one specification per moderator)
- β_1 = average treatment effect; β_2 = heterogeneity w.r.t. Z_r
- Same FE and clustering as baseline DiD (entity + year-month FE; SEs by PE)
- **Sample:** Municipalidades sector

Five moderators Z_r: non-local share, tender count, tender value, potential local firms, distance from Santiago.

Interacted DiD — Non-local Bidder Share

Outcome	$\hat{\beta}_1$	SE_1	$\hat{\beta}_2$	SE_2	N	N_{ent}
N bidders	0.9694^{***}	(0.1113)	-0.0371	(0.0583)	234,893	466
N local bidders	0.4020^{***}	(0.0350)	-0.1198^{**}	(0.0478)	234,893	466
N non-local bidders	0.6773^{***}	(0.0909)	0.0986^{**}	(0.0385)	234,893	466
N SME bidders	1.6674^{***}	(0.0801)	-0.0098	(0.0482)	234,893	466
N large bidders	-0.5881^{***}	(0.0467)	-0.0114	(0.0272)	234,893	466
Pr(single-bidder)	-0.1022^{***}	(0.0091)	-0.0112^{***}	(0.0034)	234,893	466
Pr(winner: SME)	0.1057^{***}	(0.0071)	-0.0005	(0.0041)	145,445	462
Pr(winner: large)	-0.1606^{***}	(0.0098)	-0.0081^{*}	(0.0048)	145,445	462
Pr(winner: SME & local)	0.0378^{***}	(0.0107)	-0.0068	(0.0058)	145,444	462
Pr(winner: large & local)	-0.0335^{***}	(0.0047)	0.0219^{***}	(0.0033)	145,444	462
log win bid / ref price	0.7816^{***}	(0.0517)	0.0058	(0.0279)	179,165	466
log bid / ref price	0.7644^{***}	(0.0496)	0.0560^{**}	(0.0250)	1,168,113	466

Interacted DiD — Potential Local Firms

Outcome	$\hat{\beta}_1$	SE_1	$\hat{\beta}_2$	SE_2	N	N_{ent}
N bidders	0.9619***	(0.1139)	0.0358	(0.0570)	234,893	466
N local bidders	0.3806***	(0.0379)	0.1111**	(0.0475)	234,893	466
N non-local bidders	0.6968***	(0.0933)	−0.0946***	(0.0367)	234,893	466
N SME bidders	1.6597***	(0.0822)	0.0191	(0.0476)	234,893	466
N large bidders	−0.5822***	(0.0480)	−0.0026	(0.0256)	234,893	466
Pr(single-bidder)	−0.1054***	(0.0093)	0.0123***	(0.0031)	234,893	466
Pr(winner: SME)	0.1062***	(0.0072)	−0.0005	(0.0040)	145,445	462
Pr(winner: large)	−0.1623***	(0.0101)	0.0079*	(0.0045)	145,445	462
Pr(winner: SME & local)	0.0380***	(0.0111)	0.0041	(0.0059)	145,444	462
Pr(winner: large & local)	−0.0292***	(0.0048)	−0.0212***	(0.0032)	145,444	462
log win bid / ref price	0.7825***	(0.0528)	−0.0052	(0.0272)	179,165	466
log bid / ref price	0.7704***	(0.0511)	−0.0456*	(0.0241)	1,168,113	466

Interacted DiD — Distance from Santiago

Outcome	$\hat{\beta}_1$	SE_1	$\hat{\beta}_2$	SE_2	N	N_{ent}
N bidders	0.8906***	(0.1315)	−0.0488	(0.0642)	202,070	405
N local bidders	0.4299***	(0.0459)	−0.1000*	(0.0525)	202,070	405
N non-local bidders	0.5661***	(0.1051)	0.0711*	(0.0409)	202,070	405
N SME bidders	1.5896***	(0.0919)	−0.0462	(0.0536)	202,070	405
N large bidders	−0.5937***	(0.0560)	0.0173	(0.0284)	202,070	405
Pr(single-bidder)	−0.0992***	(0.0095)	−0.0124***	(0.0035)	202,070	405
Pr(winner: SME)	0.0964***	(0.0078)	−0.0014	(0.0045)	126,484	399
Pr(winner: large)	−0.1527***	(0.0111)	−0.0053	(0.0051)	126,484	399
Pr(winner: SME & local)	0.0392***	(0.0122)	−0.0027	(0.0068)	126,483	399
Pr(winner: large & local)	−0.0283***	(0.0056)	0.0252***	(0.0036)	126,483	399
log win bid / ref price	0.7668***	(0.0596)	−0.0142	(0.0313)	154,251	404
log bid / ref price	0.7188***	(0.0548)	0.0332	(0.0282)	1,024,649	407

Interacted DiD — Total Value of Procurement

Outcome	$\hat{\beta}_1$	SE_1	$\hat{\beta}_2$	SE_2	N	N_{ent}
N bidders	0.9440***	(0.1173)	0.0579	(0.0664)	234,893	466
N local bidders	0.3494***	(0.0407)	0.1440***	(0.0524)	234,893	466
N non-local bidders	0.7138***	(0.0977)	−0.1086**	(0.0463)	234,893	466
N SME bidders	1.6493***	(0.0864)	0.0321	(0.0554)	234,893	466
N large bidders	−0.5861***	(0.0497)	0.0034	(0.0308)	234,893	466
Pr(single-bidder)	−0.1073***	(0.0098)	0.0137***	(0.0041)	234,893	466
Pr(winner: SME)	0.1088***	(0.0075)	−0.0040	(0.0047)	145,445	462
Pr(winner: large)	−0.1668***	(0.0105)	0.0135**	(0.0056)	145,445	462
Pr(winner: SME & local)	0.0397***	(0.0120)	0.0014	(0.0073)	145,444	462
Pr(winner: large & local)	−0.0265***	(0.0051)	−0.0229***	(0.0039)	145,444	462
log win bid / ref price	0.7930***	(0.0550)	−0.0202	(0.0318)	179,165	466
log bid / ref price	0.7861***	(0.0556)	−0.0616**	(0.0307)	1,168,113	466

Cross-Market Spillovers

Under construction...

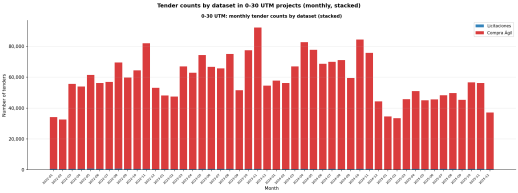
Structural Model

Under construction...

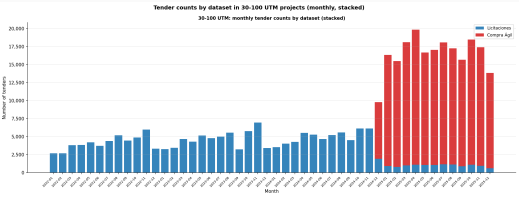
APPENDIX

App: Monthly Project Counts by Band

0–30 UTM

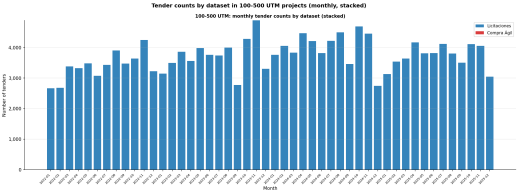


30–100 UTM (Treated Band)

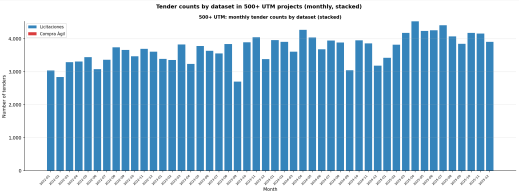


App: Monthly Project Counts — Higher Bands

100–500 UTM

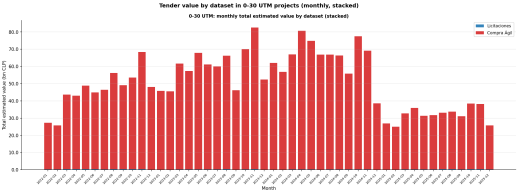


500+ UTM

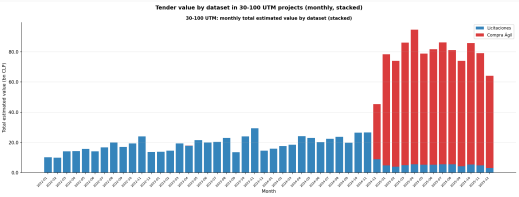


App: Monthly Project Value by Band

0–30 UTM

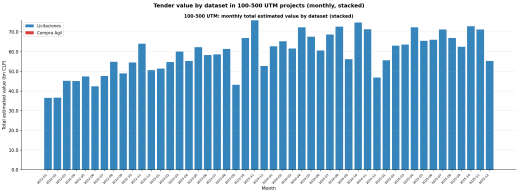


30–100 UTM (Treated Band)

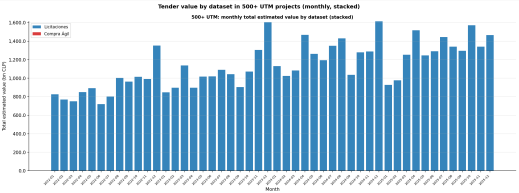


App: Monthly Project Value — Higher Bands

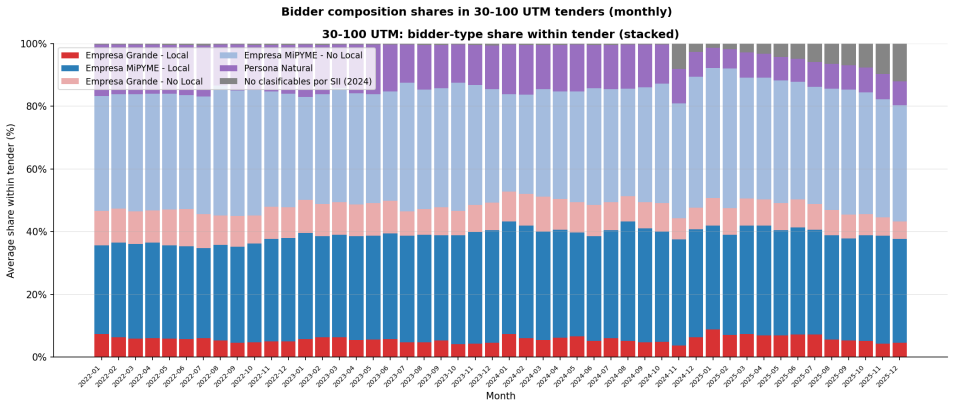
100–500 UTM



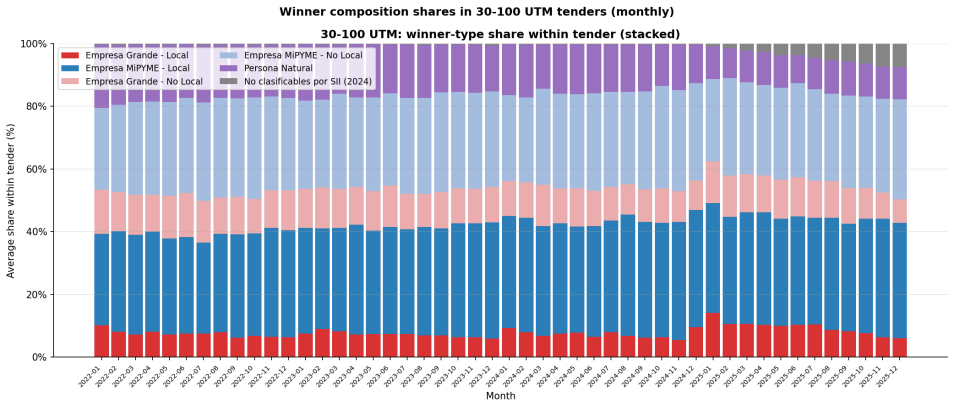
500+ UTM



App: Bidder Share by Band (30–100 UTM)



App: Winner Share by Band (30–100 UTM)



App: OLS DiD — Entry (Panel A, Municipalidades)

Outcome	Control: 0–30 UTM $\hat{\beta}_{\text{DiD}}$	N	Control: 100–200 UTM $\hat{\beta}_{\text{DiD}}$	N
Panel A: Entry at Bidding Stage				
N bidders	-0.1434^{***}	1,361,654	0.8708^{***}	234,893
	(0.0448)		(0.0973)	
N local bidders	-0.1234^{***}	1,361,654	0.3962^{***}	234,893
	(0.0211)		(0.0378)	
N non-local bidders	0.0278	1,361,654	0.5666^{***}	234,893
	(0.0349)		(0.0778)	
N SME bidders	-0.2311^{***}	1,361,654	1.4797^{***}	234,893
	(0.0439)		(0.0712)	
N large firm bidders	0.1356^{***}	1,361,654	-0.5169^{***}	234,893
	(0.0224)		(0.0396)	
N non-SII bidders	-0.0478^{***}	1,361,654	-0.0920^{**}	234,893
	(0.0084)		(0.0428)	
Pr(single-bidder tender)	0.0292^{***}	1,361,654	-0.0868^{***}	234,893
	(0.0029)		(0.0076)	

SE clustered by procuring entity. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.10$.

App: OLS DiD — Bidder Composition (Panel B, Municipalidades)

Outcome	Control: 0–30 UTM $\hat{\beta}_{\text{DiD}}$	N	Control: 100–200 UTM $\hat{\beta}_{\text{DiD}}$	N
Panel B: Bidder Composition				
Any SME bidder	-0.0116^{***}	1,283,004	0.0265^{***}	217,461
	(0.0012)		(0.0031)	
% bidders: SME	-0.0265^{***}	1,283,004	0.1024^{***}	217,461
	(0.0030)		(0.0048)	
% bidders: large firm	0.0258^{***}	1,283,004	-0.0958^{***}	217,461
	(0.0030)		(0.0047)	
% bidders: not in SII	-0.0056^{***}	1,361,654	-0.0966^{***}	234,893
	(0.0021)		(0.0059)	
% bidders: SME $\times$ local	-0.0139^{***}	1,283,004	-0.0182^{**}	217,461
	(0.0040)		(0.0072)	
% bidders: large $\times$ local	0.0077^{***}	1,283,004	-0.0277^{***}	217,461
	(0.0023)		(0.0031)	
% bidders: non-SII $\times$ local	— — —	—	— — —	—
	— — —		— — —	

SE clustered by procuring entity. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.10$.

App: OLS DiD — Winner Characteristics (Panel C, Municipalidades)

	Control: 0–30 UTM		Control: 100–200 UTM	
Outcome	$\hat{\beta}_{\text{DiD}}$	N	$\hat{\beta}_{\text{DiD}}$	N
Panel C: Winner Characteristics				
Pr(winner: SME)	-0.0436^{***}	785,164	0.0912^{***}	145,445
	(0.0051)		(0.0063)	
Pr(winner: large firm)	0.0542^{***}	785,164	-0.1353^{***}	145,445
	(0.0062)		(0.0082)	
Pr(winner: non-SII)	-0.0210^{***}	986,480	0.0008	177,903
	(0.0036)		(0.0068)	
Pr(winner: SME & local)	-0.0252^{***}	785,163	0.0350^{***}	145,444
	(0.0059)		(0.0090)	
Pr(winner: large & local)	0.0186^{***}	785,163	-0.0368^{***}	145,444
	(0.0051)		(0.0046)	
Pr(winner: non-SII & local)	— — —	—	— — —	—
	— — —		— — —	

SE clustered by procuring entity. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.10$.

App: OLS DiD — Bid Outcomes (Panel D, Municipalidades)

	Control: 0–30 UTM		Control: 100–200 UTM	
Outcome	$\hat{\beta}_{\text{DiD}}$	N	$\hat{\beta}_{\text{DiD}}$	N
Panel D: Bid Outcomes				
log(win bid/ref)	-0.0968^{***}	987,545	0.6748^{***}	179,165
	(0.0137)		(0.0444)	
log(min bid/ref)	-0.5160^{***}	1,360,213	0.9736^{***}	232,667
	(0.0195)		(0.0683)	
log(bid/ref)	-0.2937^{***}	7,197,226	0.6599^{***}	1,168,113
	(0.0107)		(0.0425)	
Bid CV	-0.0587^{***}	1,114,740	-0.1132^{***}	183,756
	(0.0039)		(0.0074)	

SE clustered by procuring entity. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.10$.

App: Bunching Tests — Motivation

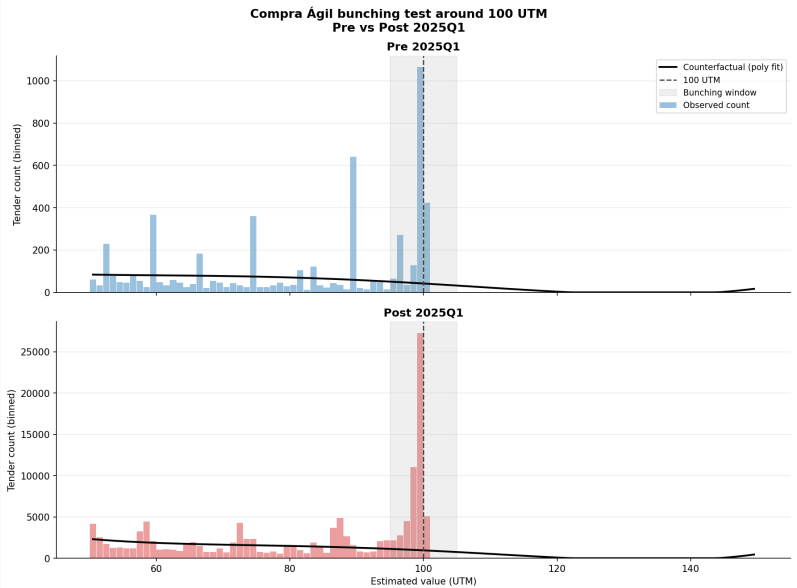
What are we testing? If agencies strategically size contracts to stay below a UTM threshold, we should observe an excess density of contracts just below that threshold.

Method: Fit a 4th-order polynomial to the contract-value histogram, excluding a ± 5 UTM window around the threshold. Excess bunching ratio $=$ (actual $-$ counterfactual)/counterfactual in the window. A band test compares the below/above-threshold count ratio pre- vs. post-reform using a χ^2 test.

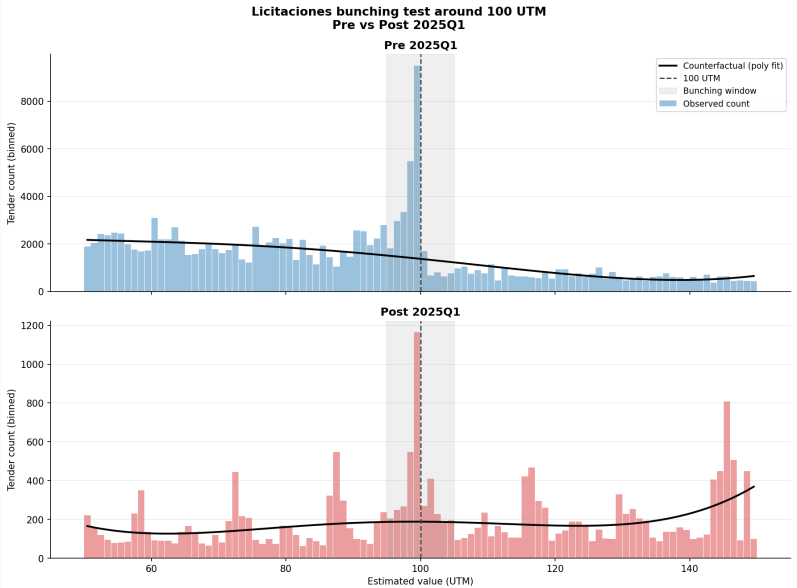
Two thresholds tested:

- **100 UTM** (main): the local preference threshold introduced by the Compra Ágil reform
- **30 UTM** (placebo): the prior Compra Ágil ceiling, now an interior point of the treated band

App: Bunching — Compra Ágil at 100 UTM

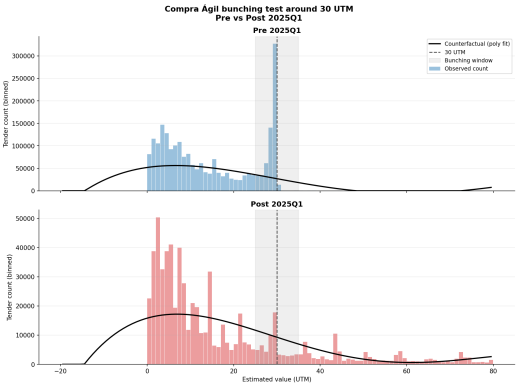


App: Bunching — Licitaciones at 100 UTM (Placebo)

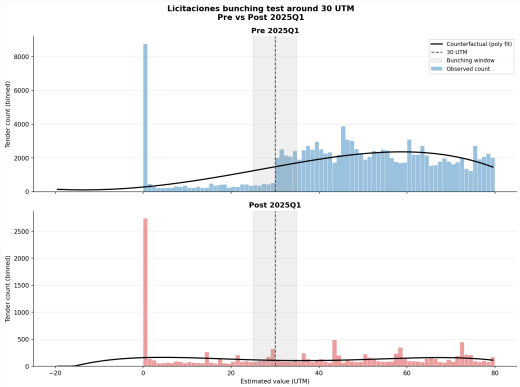


App: Bunching at 30 UTM (Old Threshold)

Compra Ágil



Licitaciones



Pre-reform: massive below-threshold bunching in Compra Ágil (below/above = 42.8). Post-reform: collapses to 2.79 ($\chi^2 = 74{,}299$, $p < 0.001$). Bunching migrates from 30 UTM to 100 UTM after the reform.

App: Bunching — Band Test Summary

Threshold	Contract type	Pre below/above	Post below/above	χ^2	p-value
100 UTM	Compra Ágil	3.70	9.43	289.1	< 0.001
100 UTM	Licitaciones	5.05	1.88	710.5	< 0.001
30 UTM	Compra Ágil	42.8	2.79	74,299	< 0.001
30 UTM	Licitaciones	0.19	1.69	1,530	< 0.001

All tests compare the below/above ratio pre- vs. post-2025Q1. The Compra Ágil 30 UTM result is the sharpest signal: extreme pre-reform bunching at the old ceiling disappears after the threshold migrates to 100 UTM.